

HMIC 2020 Solutions

April 21, 2020

1. [6] Sir Alex is coaching a soccer team of n players of distinct heights. He wants to line them up so that for each player P , the total number of players that are either to the left of P and taller than P or to the right of P and shorter than P is even. In terms of n , how many possible orders are there?

Proposed by: Michael Ren

Answer: $\boxed{\lfloor \frac{n}{2} \rfloor! \lceil \frac{n}{2} \rceil!}$

Solution: We want to determine the number of permutations π of $1, \dots, n$ such that every number is involved in an even number of inversions. The key observation is that the number of inversions k is involved in has the same parity as $\pi(k) - k$. This is since $\pi(k) - k$ can be interpreted as the number of i with $\pi(i) < \pi(k)$ minus the number of i with $i < k$, meaning that i is counted with a weight of $\pm 1 \equiv 1 \pmod{2}$ if and only if i is in an inversion with k .

From this, it follows that such permutations are the ones that permute the even and odd numbers, so the answer is $\lfloor \frac{n}{2} \rfloor! \lceil \frac{n}{2} \rceil!$.

2. [7] Some bishops and knights are placed on an infinite chessboard, where each square has side length 1 unit. Suppose that the following conditions hold:

- For each bishop, there exists a knight on the same diagonal as that bishop (there may be another piece between the bishop and the knight).
- For each knight, there exists a bishop that is exactly $\sqrt{5}$ units away from it.
- If any piece is removed from the board, then at least one of the two conditions above is no longer satisfied.

If n is the total number of pieces on the board, find all possible values of n .

Proposed by: Sheldon Kieren Tan

Answer: $\boxed{4 \mid n}$

Solution: Color the chessboard with the usual chessboard coloring. Note that bishops can only attack knights on the same colored squares, while knights can only attack bishops on different colored squares. Let B_B, W_B, B_N, W_N denote the number of bishops on black colored squares, bishops on white colored squares, knights on black colored squares and knights on white colored squares respectively.

Since removing a knight on a white colored square will cause a leave a bishop on a white colored square with no knight to attack, $W_N \leq W_B$. Similarly, $B_N \leq B_B$, $B_B \leq W_N$, $W_B \leq B_N$. Combining the inequalities, we find that $W_N = W_B = B_N = B_B$. Hence $4 \mid n$.

To construct $n = 4$, we can place bishops at $(1, 0), (2, 2)$ and knights at $(0, 1), (3, 1)$. We can easily extend this to any multiple of 4 by placing this configuration as many times as required across the grid, making sure that we avoid the (finitely many) diagonals that existing bishops occupy.

(A harder variant of the problem involves changing the third condition from "If any piece" to "If any proper subset of pieces". The proof remains unchanged but the construction is more tedious as the pieces must be "connected" without accidentally attacking one another. The question can also be phrased in a non-chess way to obscure the motivation behind the coloring.)

3. [9] Let $P_1 P_2 P_3 P_4$ be a tetrahedron in $\mathbb{R}^3$ and let O be a point equidistant from each of its vertices. Suppose there exists a point H such that for each i , the line $P_i H$ is perpendicular to the plane through the other three vertices. Line $P_1 H$ intersects the plane through P_2, P_3, P_4 at A , and contains a point $B \neq P_1$ such that $OP_1 = OB$. Show that $HB = 3HA$.

Proposed by: Michael Ren

Solution 1: Note that A is the orthocenter of triangle $P_2P_3P_4$ by projecting the givens about H .

Next, let P_2A intersect P_3P_4 at C and $(P_2P_3P_4)$ at D . Then $AD = 2AC$.

Note that H is the orthocenter of triangle P_1P_2C , with A being the foot of the altitude from P_1 . Thus $AP_2 \cdot AC = AH \cdot AP_1$.

Finally, by Power of a Point at A ,

$$AP_1 \cdot AB = AP_2 \cdot AD = 2AP_2 \cdot AC = 2AH \cdot AP_1.$$

Thus $AB = 2AH$ which gives $HB = 3HA$, as desired.

Solution 2: We use vectors with H as the origin. Then $P_i \cdot (P_j - P_k) = 0$ from the definition of H . Thus $P_i \cdot P_j = P_i \cdot P_k$ for any distinct i, j, k . In particular, the quantity $P_i \cdot P_j$ is a constant over all distinct choices of i, j . Let this value be D . Let

$$O' = \frac{1}{2}(P_1 + P_2 + P_3 + P_4).$$

We claim $O' = O$. To see this, note that

$$(O' - P_1) \cdot (O' - P_1) = \frac{1}{4} (P_2 + P_3 + P_4 - P_1) \cdot (P_2 + P_3 + P_4 - P_1)$$

will be symmetric after expanding. Thus O' is equidistant from the P_i 's, which means $O' = O$.

Now, the projection of O onto P_1H is $\frac{O \cdot P_1}{P_1 \cdot P_1} P_1$, so

$$B = P_1 \left(\frac{(2O - P_1) \cdot P_1}{P_1 \cdot P_1} \right) = P_1 \left(\frac{3D}{P_1 \cdot P_1} \right).$$

Meanwhile, A is the projection of P_2 on P_1H , so

$$A = P_1 \left(\frac{P_1 \cdot P_2}{P_1 \cdot P_1} \right) = P_1 \left(\frac{D}{P_1 \cdot P_1} \right).$$

Thus $HB = 3HA$, as desired.

4. [9] Let $C_k = \frac{1}{k+1} \binom{2k}{k}$ denote the k^{th} Catalan number and p be an odd prime. Prove that exactly half of the numbers in the set

$$\left\{ \sum_{k=1}^{p-1} C_k n^k \mid n \in \{1, 2, \dots, p-1\} \right\}$$

are divisible by p .

Proposed by: Tristan Shin

Solution 1: We work in $\mathbb{F}_p[X]$.

We claim that

$$(1 - 4X)^{\frac{p+1}{2}} - 1 + 2X^p + 2X \sum_{k=0}^{p-1} C_k X^k = 0$$

The solution follows from this claim, as

$$\sum_{k=1}^{p-1} C_k n^k \equiv 0 \pmod{p} \iff (1 - 4n)^{\frac{p+1}{2}} - 1 + 2n^p + 2n \equiv 0 \pmod{p}.$$

Since $n^p \equiv n \pmod{p}$, this is equivalent to $n \equiv \frac{1}{4} \pmod{p}$ or $1 - 4n$ being a non-zero quadratic residue in $\mathbb{F}_p$. But we must omit the solution of $n = 0$, so we get $1 + (\frac{p-1}{2} - 1) = \frac{p-1}{2}$ values of n that work.

Now we prove the claim. Observe the following facts:

- $C_{p-1} = \frac{1}{2^{p-1}} \binom{2p-1}{p} = -1$ by Lucas' theorem.
- For $\frac{p-1}{2} < k < p-1$,

$$\nu_p(C_k) = \nu_p((2k)!) - \nu_p((k+1)!) - \nu_p(k!) = 1$$

since $k+1 < p \leq 2k$, so $C_k = 0$.

We show that the coefficient of X^k in the LHS is 0 for $k = 0, \dots, p$. This is obvious for $k > \frac{p+1}{2}$ and $k = 0$ by the above facts, as the coefficient is $2 + 2C_{p-1}$ for $k = p$, $2C_k$ for $\frac{p+1}{2} < k < p$, and 0 for $k = 0$.

Now for $1 \leq k \leq \frac{p+1}{2}$, the coefficient is $\binom{\frac{p+1}{2}}{k}(-4)^k + 2C_{k-1} = 0$. Write

$$\begin{aligned} \binom{\frac{p+1}{2}}{k} &= \frac{\frac{p+1}{2} \cdot \frac{p-1}{2} \cdot \frac{p-3}{2} \dots \frac{p-2k+3}{2}}{k!} \\ &= \frac{\frac{1}{2} \cdot (-\frac{1}{2}) \cdot (-\frac{3}{2}) \dots (-\frac{2k-3}{2})}{k!} \\ &= \frac{(-1)^{k-1}}{2^k} \cdot \frac{1 \cdot 3 \dots (2k-3)}{k!} \\ &= \frac{(-1)^{k-1}}{2^k} \cdot \frac{(2k-2)!}{2^{k-1}(k-1)!k!} \\ &= -\frac{2C_{k-1}}{(-4)^k} \end{aligned}$$

so the coefficient is 0.

Solution 2: We present an alternate proof of the key claim. Use the same starting facts as before.

$$\text{Let } Q(X) = -2 + \sum_{k=0}^{\frac{p-1}{2}} C_k X^k = -2 + X^{p-1} + \sum_{k=0}^{p-1} C_k X^k.$$

Square Q and multiply by X to get

$$\begin{aligned} XQ^2 &= 4X - 4X \sum_{k=0}^{\frac{p-1}{2}} C_k X^k + X \left(\sum_{k=0}^{\frac{p-1}{2}} C_k X^k \right)^2 \\ &= -4X - 4XQ + X \sum_{k=0}^{p-2} \sum_{j=0}^k C_j C_{k-j} X^k + C_{\frac{p-1}{2}}^2 X^p \\ &= -4X - 4XQ + \sum_{k=0}^{p-2} C_{k+1} X^{k+1} + 4X^p \\ &= -4X - 4XQ + Q + 1 - X^{p-1} + 4X^p \\ &= (1 - 4X)Q - (1 - 4X)(X^{p-1} - 1). \end{aligned}$$

It follows that

$$\begin{aligned} 0 &= 4X \cdot (XQ^2 - (1 - 4X)Q + (1 - 4X) \cdot (X^{p-1} - 1)) \\ &= 4X^2 Q^2 - 4X(1 - 4X)Q + 4(1 - 4X) \cdot (X^p - X) \\ &= 4X^2 Q^2 - 4X(1 - 4X)Q + (1 - 4X)^2 - (1 - 4X)^{p+1} \\ &= (2XQ + (1 - 4X)^{\frac{p+1}{2}} - (1 - 4X)) \cdot (2XQ - (1 - 4X)^{\frac{p+1}{2}} - (1 - 4X)). \end{aligned}$$

Evaluating the second factor at 0 gives -2 , so it is not the zero polynomial. Thus the first factor is the zero polynomial, from which the claim follows.

Solution 3: We prove the following generalization: Let q be a power of p . Then the polynomial $\sum_{k=1}^{q-1} C_k X^k$ has $\frac{q+1}{2}$ roots in $\mathbb{F}_q$ and $\frac{p+1}{2}$ roots in $\mathbb{F}_p$. It once again suffices to prove the key claim, just with p replaced by q .

Work in $\mathbb{F}_q[[X]]$, the ring of formal power series over $\mathbb{F}_q$. Then

$$\begin{aligned} \sum_{k=0}^{\infty} \binom{2k}{k} X^k &= (1 - 4X)^{-\frac{1}{2}} \\ &= (1 - 4X)^{\frac{q-1}{2}} ((1 - 4X)^q)^{-\frac{1}{2}} \\ &= (1 - 4X)^{\frac{q-1}{2}} (1 - 4X^q)^{-\frac{1}{2}} \\ &= (1 - 4X)^{\frac{q-1}{2}} \sum_{k=0}^{\infty} \binom{2k}{k} X^{qk}. \end{aligned}$$

Taking both sides mod X^q ,

$$\sum_{k=0}^{q-1} \binom{2k}{k} X^k = (1 - 4X)^{\frac{q-1}{2}}.$$

Then using $\binom{2q}{q} = 2$, we have

$$\sqrt{1 - 4X} = (1 - 4X) \cdot (1 - 4X)^{-\frac{1}{2}} \equiv (1 - 4X) \cdot ((1 - 4X)^{\frac{q-1}{2}} + 2X^q) \pmod{X^{q+1}}$$

so

$$X \sum_{k=0}^{\infty} C_k X^k = \frac{1 - \sqrt{1 - 4X}}{2} \equiv \frac{1 - (1 - 4X)^{\frac{q+1}{2}} + 2X^q}{2} \pmod{X^{q+1}}$$

from which the claim follows.

Alternatively, one can also finish by integrating $(1 - 4X)^{\frac{p-1}{2}}$, noting that the “constant of integration” is no longer a constant but rather a polynomial of the form $a + bX^p$.

5. [11] A triangle and a circle are in the same plane. Show that the area of the intersection of the triangle and the circle is at most one third of the area of the triangle plus one half of the area of the circle.

Proposed by: Krit Boonsiriseth

Solution 1: (Ankan Bhattacharya¹) Let $\triangle$ denote the triangle, $\circ$ denote the circle, and $\triangle'$ denote the reflection of the triangle in the center of the circle. Letting overline denote complement,

$$\begin{aligned} [\triangle \cap \circ] &\leq \frac{1}{3}[\triangle] + \frac{1}{2}[\circ] \\ \iff [\triangle \cap \triangle' \cap \circ] + [\triangle \cap \overline{\triangle'} \cap \circ] &\leq \frac{1}{3}[\triangle] + \frac{1}{2}[[\circ \cap \triangle \cap \triangle'] + 2[\circ \cap \triangle \cap \overline{\triangle'}] + [\circ \cap \overline{\triangle} \cap \overline{\triangle'}]] \\ \iff \frac{1}{2}[\triangle \cap \triangle' \cap \circ] &\leq \frac{1}{3}[\triangle] + \frac{1}{2}[\circ \cap \overline{\triangle} \cap \overline{\triangle'}] \\ \iff [\triangle \cap \triangle' \cap \circ] &\leq \frac{2}{3}[\triangle] + [\circ \cap \overline{\triangle} \cap \overline{\triangle'}]. \end{aligned}$$

Since $\triangle \cap \triangle' \cap \circ$ is centrally symmetric, it's enough to show the following well-known lemma.

Lemma. A centrally symmetric region $\mathcal{R}$ of a triangle can have area at most $\frac{2}{3}$ that of the triangle.

Let the triangle be ABC with medial triangle DEF . We take two cases.

¹<https://artofproblemsolving.com/community/c6h2052367p14590886>

- **Case 1.** The center of symmetry is in one of the outer triangles, say $\triangle AEF$. Then the maximal possible $\mathcal{R}$ is a parallelogram with one vertex A and two other vertices on $\overline{AB}$ and $\overline{AC}$. By enlargening the parallelogram, we may assume its fourth vertex lies on $\overline{BC}$. If this vertex divides $\overline{BC}$ into an $a : b$ ratio with $a + b = 1$, then the fraction of the area taken up by the parallelogram is $1 - a^2 - b^2 \leq \frac{1}{2}$.
- **Case 2.** The center of symmetry is in $\triangle DEF$. Then the maximal possible $\mathcal{R}$ is a centrally symmetric hexagon with two vertices on each side. Then there are three little triangles similar to $\triangle ABC$ on the hexagon; by equal lengths, the similarity ratios a, b, c sum to 1. Then the fraction of the area taken up by the hexagon is $1 - a^2 - b^2 - c^2 \leq \frac{2}{3}$. $\square$

We are done.

Solution 2: It is also possible to approach this as an optimization problem. Fix a triangle $\triangle = ABC$ on the plane, vary the circle $\odot = \odot(O, r)$, and consider the objective function

$$f(O, r) := [\triangle \cap \odot] - \frac{1}{2}[\odot].$$

By a compactness argument we can show that f reaches a maximum, and at that maximum $\frac{\partial f}{\partial r} = 0$ and $\nabla_O f = \vec{0}$ must simultaneously hold.

- We have

$$\begin{aligned} \frac{\partial}{\partial r} f(O, r) &= \frac{\partial}{\partial r} [\triangle \cap \odot] - \frac{1}{2} \frac{\partial}{\partial r} [\odot] \\ &= r \cdot ((\text{total angle subtended by } AB \cap \odot, BC \cap \odot, CA \cap \odot \text{ at point } O) - \pi), \end{aligned}$$

therefore

$$\frac{\partial}{\partial r} f(O, r) = 0 \iff \text{the total angle subtended by the sides of } \triangle \text{ inside } \odot \text{ is } \pi. \quad (1)$$

- Let $\vec{n}_a, \vec{n}_b, \vec{n}_c$ denote the normal vectors of BC, CA and AB . Define k_a to be the length of $BC \cap \odot$, and define k_b, k_c similarly.

$$\begin{aligned} \nabla_O f(O, r) &= \nabla_O [\triangle \cap \odot] - \frac{1}{2} \nabla_O [\odot] \\ &= (k_a \vec{n}_a + k_b \vec{n}_b + k_c \vec{n}_c) - \vec{0} \end{aligned}$$

It is easy to see that $a\vec{n}_a + b\vec{n}_b + c\vec{n}_c = \vec{0}$ is the only linear relation between $\vec{n}_a, \vec{n}_b, \vec{n}_c$, so

$$\nabla_O f(O, r) = \vec{0} \iff k_a : k_b : k_c = a : b : c \quad (2)$$

Now suppose that $\odot(O, r)$ is chosen so that $\frac{\partial f}{\partial r} = 0$ and $\nabla_O f = \vec{0}$, so the angle condition in (1) holds, and $(k_a, k_b, k_c) = k(a, b, c)$ for some k . There are two possible cases:

- **Case 1.** All vertices of $\triangle$ do not lie inside $\odot$. In this case $\odot$ intersects $\triangle$ at six points. The lines joining O and these six points divide $\triangle \cap \odot$ into three sectors and three triangles. By the angle condition in (1), the three sectors have total area $\frac{1}{2}[\odot]$. The three triangles have total area $k[\triangle]$, and it suffices to show that $k \leq \frac{1}{3}$.

As the total angle of the triangles at O is π , we may join two copies of each of the three triangles to form a cyclic hexagon. Therefore,

$$\begin{aligned} 2k \cdot [\triangle] &= 2 \cdot (\text{total area of three triangles}) \\ &= \text{area of cyclic hexagon with side lengths } ka, kb, kc, ka, kb, kc \\ &\geq \text{area of hexagon with side lengths } ka, kb, kc, ka, kb, kc \\ &\quad \text{created by six copies of a triangle with side lengths } ka, kb, kc \\ &= 6k^2 \cdot [\triangle], \end{aligned}$$

and our conclusion readily follows.

- **Case 2.** A vertex of $\triangle$ lies in the interior of $\odot$. WLOG let that vertex be A . Let $\odot$ intersect AB at X , AC at Y , and BC at D and E . From (2), $AX : AY : DE = AB : AC : BC$, $XY \parallel BC$ and $XY = DE$, therefore $\square XYED$ is a rectangle. From (1), $\angle XOD + \angle YOE = \pi$, so in fact $\square XYED$ is a square. Let x be the side length of $\square XYDE$, and let h be the height of the altitude from A to BC . Clearly $h > x$. As A lies inside $\odot = (XYDE)$, $\angle BAC > \frac{3\pi}{4}$, so $a > 3h > 3x$. Now we are done because $k = \frac{x}{a} < \frac{1}{3}$. ■